AWS Elastic IP Addresses - Service Overview

Introduction

An Elastic IP address is a static, public IPv4 address designed for dynamic cloud computing, allocated to your AWS account and reachable from the internet. Unlike a standard EC2 public IP address, which changes each time an instance is stopped and started, an Elastic IP address remains associated with your account until you explicitly release it.

Why Elastic IPs Exist

When an EC2 instance is stopped and restarted, AWS releases its previous public IP address and assigns a new one. For most applications this is not an issue, since DNS or a load balancer typically fronts the instance. However, for scenarios requiring a fixed, known public IP (for example, an application whitelisted by IP at a partner organization, or a mail server needing a stable reverse-DNS-able IP), an Elastic IP solves this problem by decoupling the address from the instance's lifecycle.

Allocation and Association

An Elastic IP is first allocated to your AWS account, then associated with a specific EC2 instance or network interface. Because the address is allocated independently, it can be quickly remapped from one instance to another, for example during a failover event, without needing to update DNS records, since the IP itself does not change.

Pricing Model

Elastic IPs are free of charge while actively associated with a running instance. However, AWS charges an hourly fee for Elastic IP addresses that are allocated to your account but NOT associated with a running instance, or associated with a stopped instance. This pricing model is specifically designed to discourage hoarding of scarce IPv4 address space, since public IPv4 addresses are a limited global resource.

Limits

By default, an AWS account is limited to 5 Elastic IP addresses per region, though this limit can be raised via a service quota increase request if a legitimate use case is demonstrated.

Relationship to NAT Gateways

A NAT Gateway, which allows instances in a private subnet to initiate outbound internet traffic without being directly reachable from the internet, requires an Elastic IP address to be allocated to it at creation time. This is one of the most common sources of Elastic IP usage in a typical VPC architecture.

Relationship to Load Balancers

Application Load Balancers and Network Load Balancers typically do not use Elastic IP addresses directly; ALBs receive a DNS name backed by rotating IPs, while NLBs can optionally support Elastic IPs per Availability Zone for cases requiring a fixed IP endpoint.

IPv4 Scarcity and BYOIP

Global IPv4 address space has been fully allocated for years, which is the underlying reason AWS charges for idle Elastic IPs and imposes a default per-region quota. For organizations with their own publicly routable IPv4 address ranges (obtained from a regional internet registry), AWS supports Bring Your Own IP (BYOIP), allowing you to migrate an existing public IPv4 (or IPv6) range into your AWS account and allocate Elastic IPs from within that range, useful for preserving IP reputation when migrating a mail server or maintaining continuity with existing firewall allowlists during a cloud migration.

Elastic IP vs Public IP Recap

To summarize the distinction precisely: a standard EC2 public IP is allocated automatically, cannot be reserved in advance, and is released back to AWS's pool the moment the instance is stopped or terminated. An Elastic IP is allocated on request, persists independently of any instance's state, and must be manually released when no longer needed, at which point it returns to AWS's general pool and may be issued to a different account.

Common Use Cases

- Providing a stable public IP for a NAT Gateway
- Whitelisting a fixed IP with an external partner or third-party API
- Fast failover between a primary and standby instance without DNS propagation delay
- Hosting a small number of services that must be reachable at a known, unchanging address

Best Practices

- Release any Elastic IP address that is not actively in use, to avoid unnecessary hourly charges.
- Prefer a load balancer with a DNS name over a bare Elastic IP for applications that can tolerate DNS-based routing, since this is more resilient and scalable.
- Monitor Elastic IP usage across an account periodically, since unused allocated addresses are a

common source of unexpected AWS bill line items.